

LABORATORY WORK 1 — THE PRICE OF ONE REQUEST

LLMs, Agentic AI and Reinforcement Learning

1. Part 1 prediction and measured value

For the complaint, I predicted the token ratios from the Part 1 byte ratios: RU/EN $\approx 1.92\times$ and KK/EN $\approx 2.13\times$. The measured complaint ratios were $1.36\times$ and $2.00\times$ with o200k_base, and $2.47\times$ and $4.49\times$ with cl100k_base. Therefore, the prediction was much closer to the o200k_base result for Kazakh, while the cl100k_base tokenizer shows a much larger language premium. This demonstrates that bytes and characters do not determine tokenizer cost directly.

2. Annual cost at 2,000 support requests/day

I use 2,000 requests/day as a representative support-queue volume because it is the volume used by the supplied lab output and makes the annual cost comparison concrete.

Model	English	Russian	Kazakh
haiku-4.5	\$1,596	\$2,366	\$3,947
sonnet-5	\$3,192	\$4,732	\$7,894
opus-5	\$7,979	\$11,830	\$19,736
fable-5.1	\$15,958	\$23,659	\$39,471

3. Production model for a Kazakh support queue

I would choose sonnet-5 as a practical cost/quality balance for a Kazakh-language support queue. Its annual cost for the 2,000-request/day scenario is \$7,894, compared with \$3,947 for haiku-4.5 and \$19,736 for opus-5. The quality side should be checked with the lab's four-point checklist: do not invent a reason for the rate change, do not invent numbers, answer in the customer's language, and provide a concrete next step. The final production choice should be confirmed only after this quality check.

4. Main conclusion

The experiment shows that token cost depends on the tokenizer and on generated answer length, not simply on characters or bytes. For the reference run, the actual Opus bill ratio was $1.48\times$ for Russian versus English and $2.47\times$ for Kazakh versus English, while the input-only ratios were $1.30\times$ and $2.35\times$. Thus, input-token ratio and total-bill ratio are different measures.

5. Cost lever not used

One cost lever not used by the lab is prompt caching; batch discounts were also not applied.

HOW I USED AI IN THE LABORATORY WORK

1. Purpose I used AI to understand the task, explain tokenization concepts, and organize the report.

2. How I used AI I used AI to formulate the prediction, explain characters, bytes, words and tokens, and interpret cost differences between English, Russian and Kazakh.

3. Verification I checked the AI-generated reasoning against the actual Part 0–3 outputs. The final numbers were taken from the laboratory results.

4. Reflection AI helped me understand why Kazakh can require more tokens and how tokenization affects cost. I learned that AI can assist with analysis, but numerical results must be verified.

5. Workflow Read the assignment → make a prediction → run the measurements → compare results → calculate costs → prepare the report.